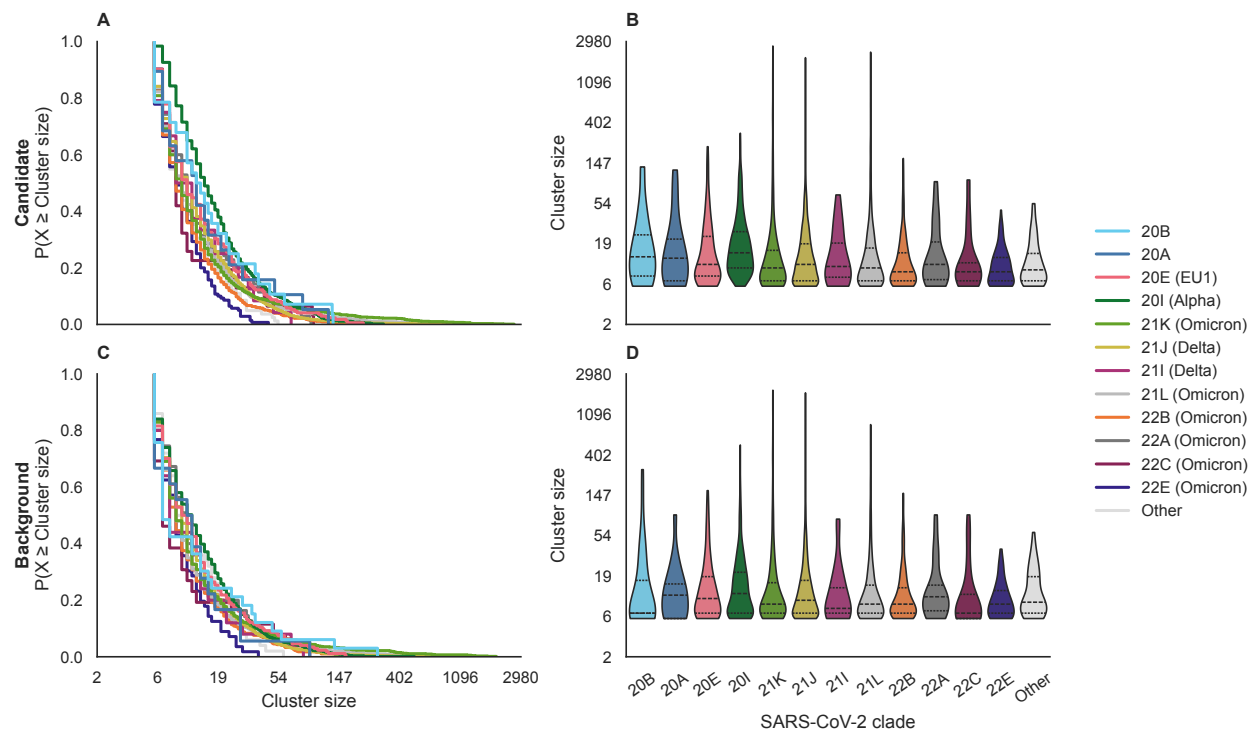


Supporting Information

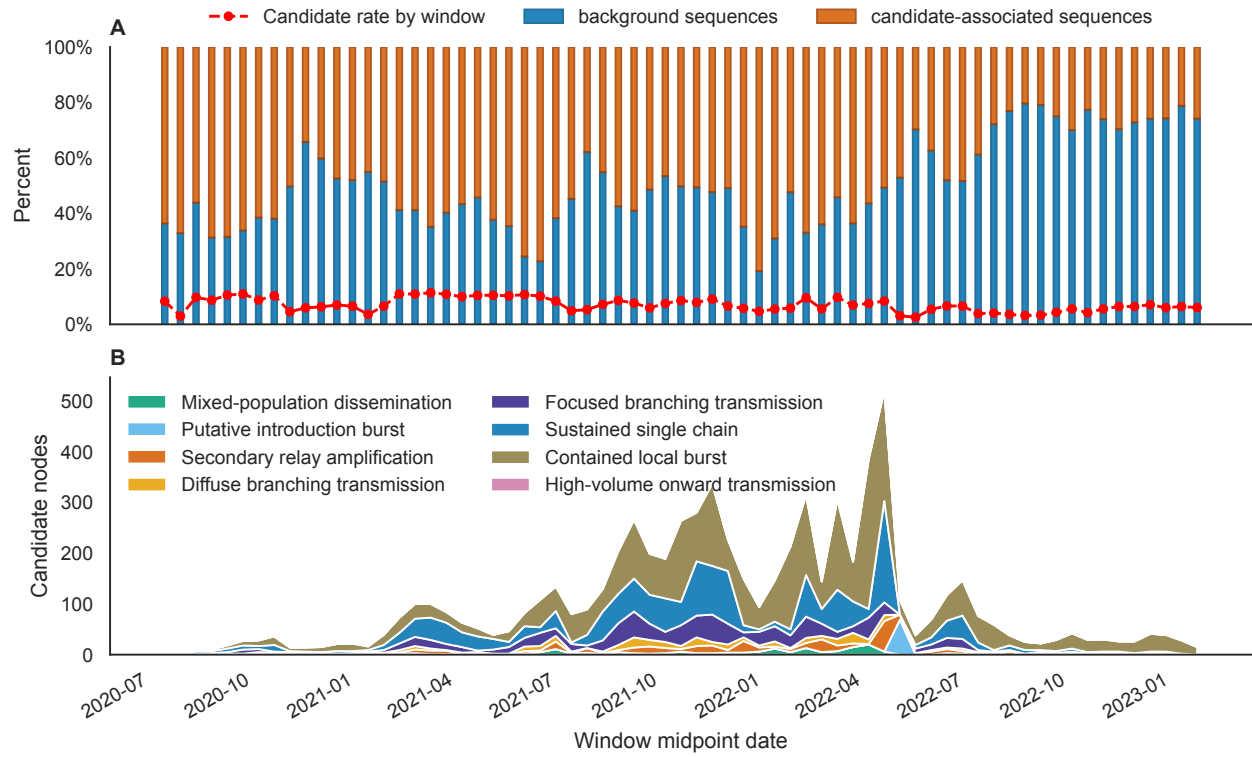
Detecting superspreading signatures from national SARS-CoV-2 genomic surveillance using EpiLink-derived temporal cluster-transition graphs

Dominic Arthur, Christopher J. Banks, Rowland R. Kao

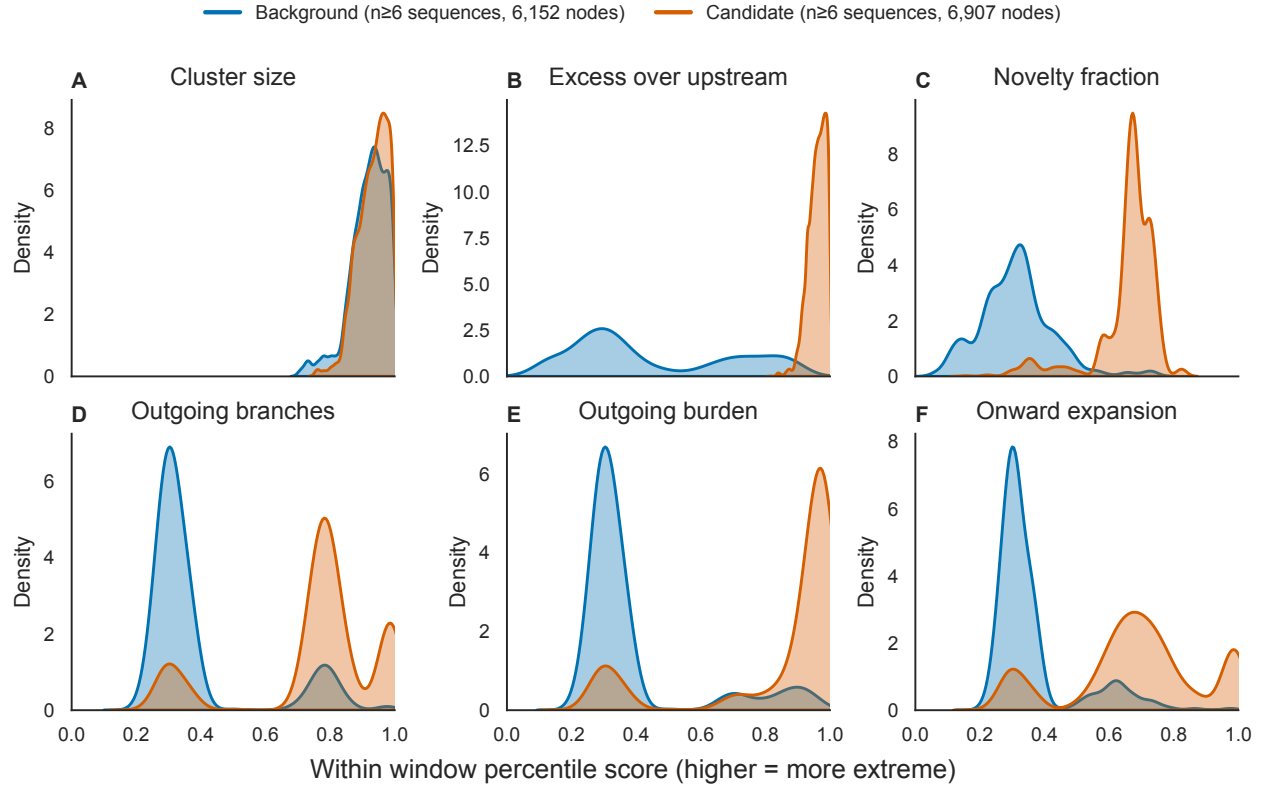
Supporting figures



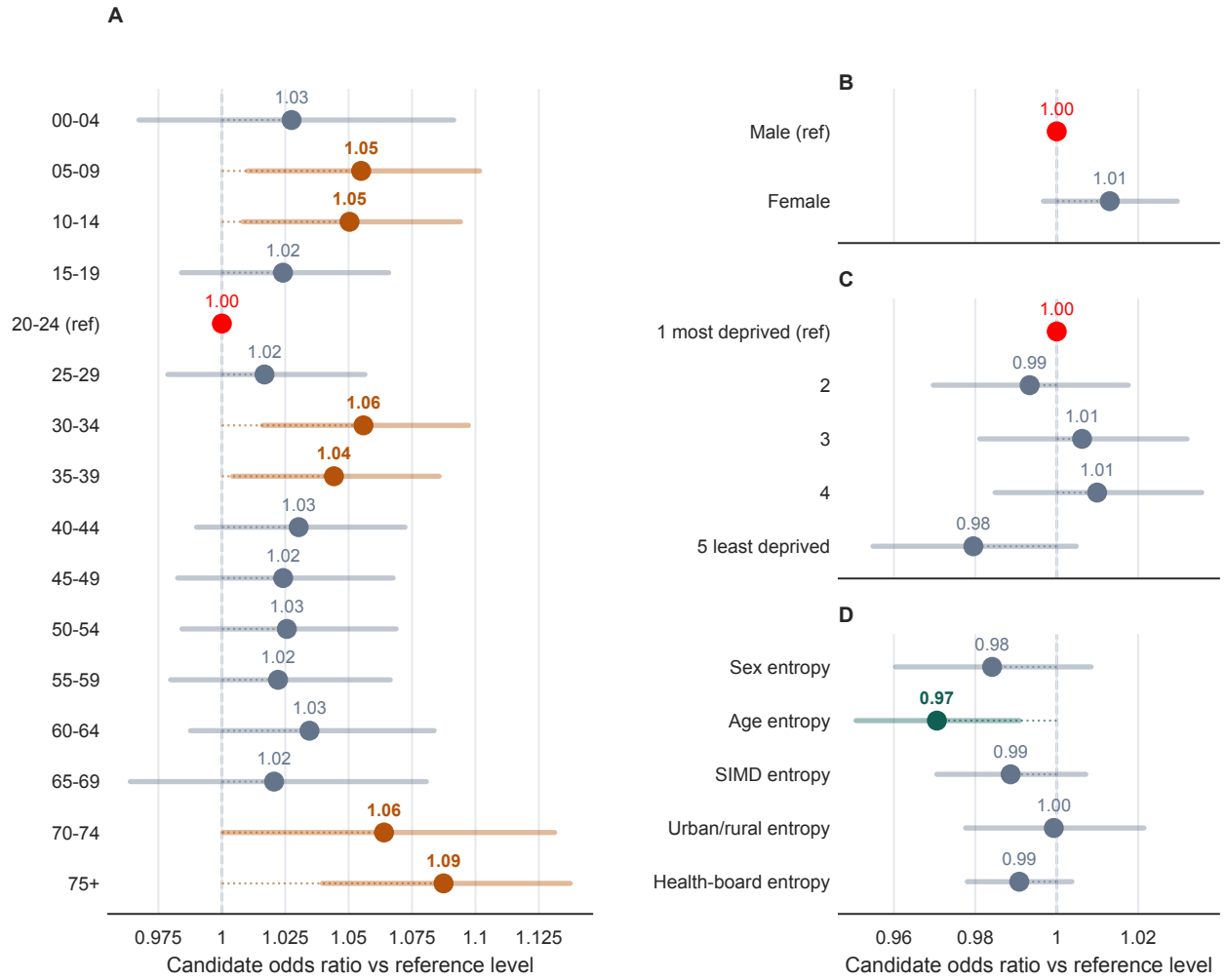
S1 Fig. Candidate cluster-size distribution. The strongly right-skewed cluster-size distribution of candidate SSE nodes.



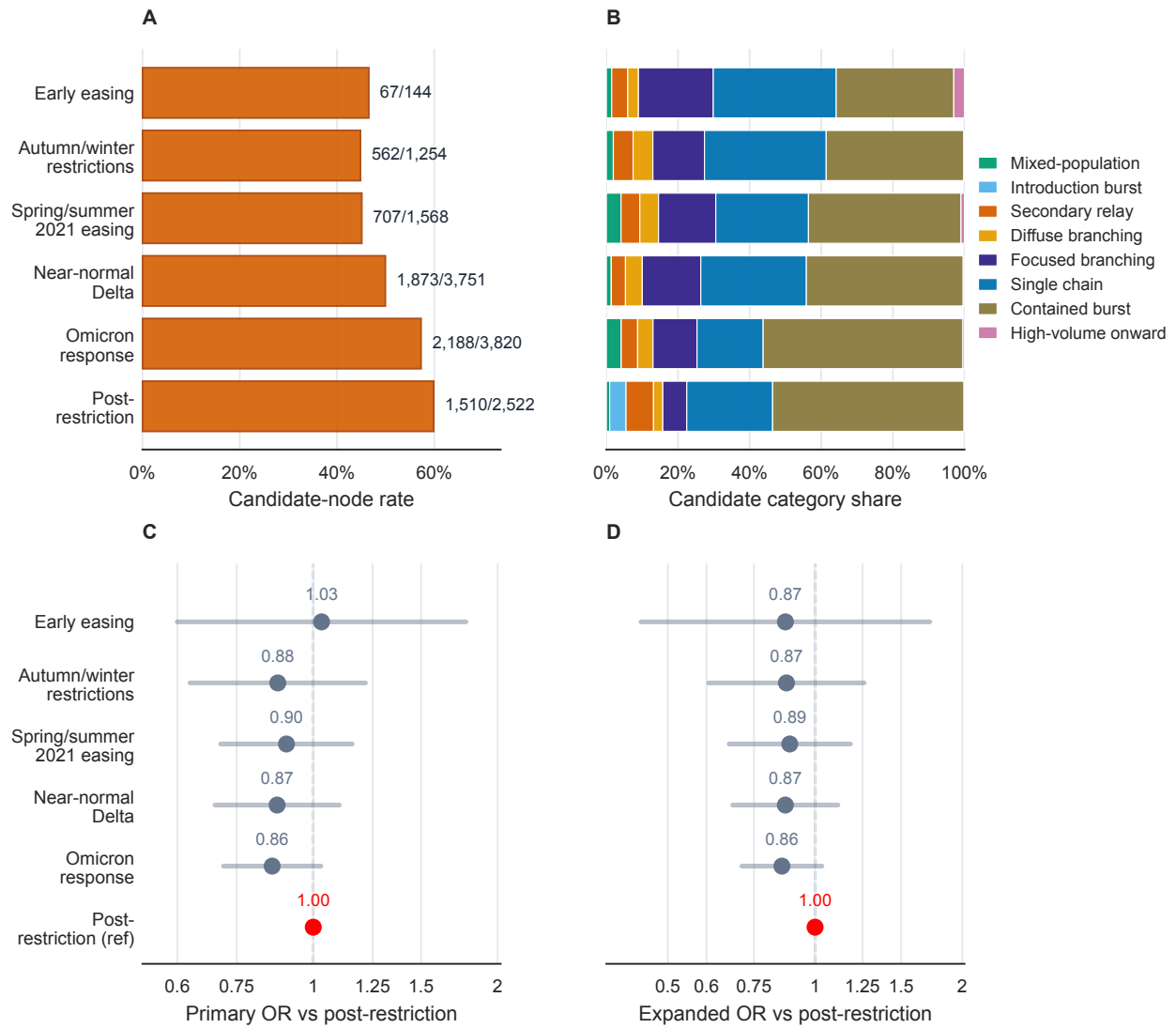
S2 Fig. Candidate-node frequency over time. Candidate share across surveillance windows and the major epidemic phases.



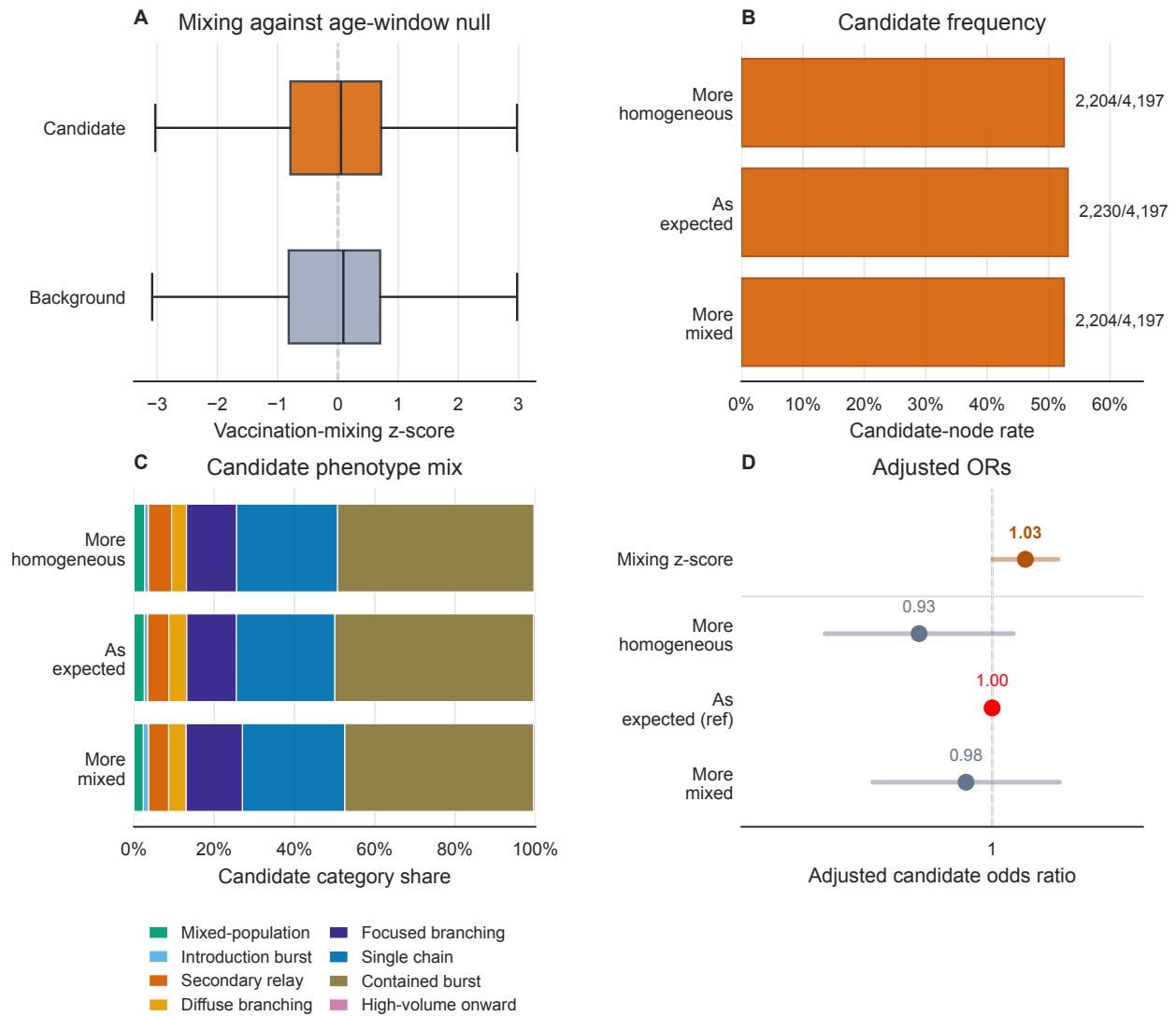
S3 Fig. Within-window score components. Overlaid candidate versus size-matched background distributions of the six within-window percentile components underlying the candidate screen (cluster size, excess over upstream continuity, novelty fraction, outgoing branches, outgoing burden, onward expansion).



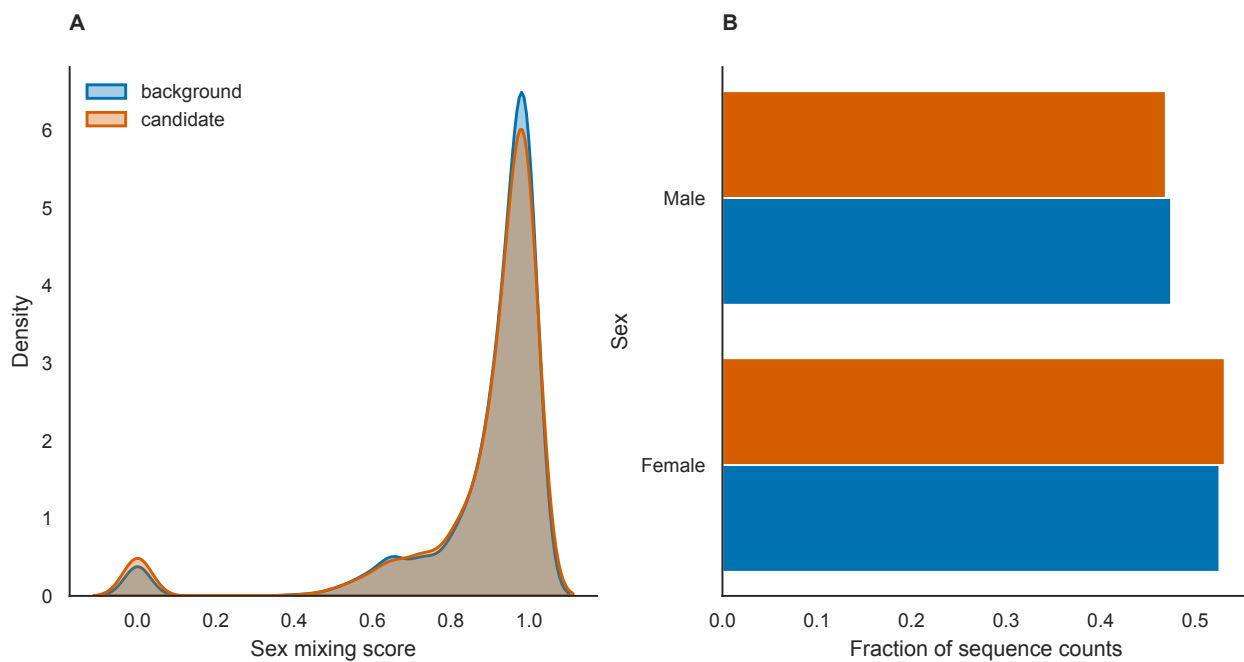
S4 Fig. Individual demographic and node-mixing odds ratios. Forest plots of individual demographic and entropy-based node-mixing associations with candidate-node membership.



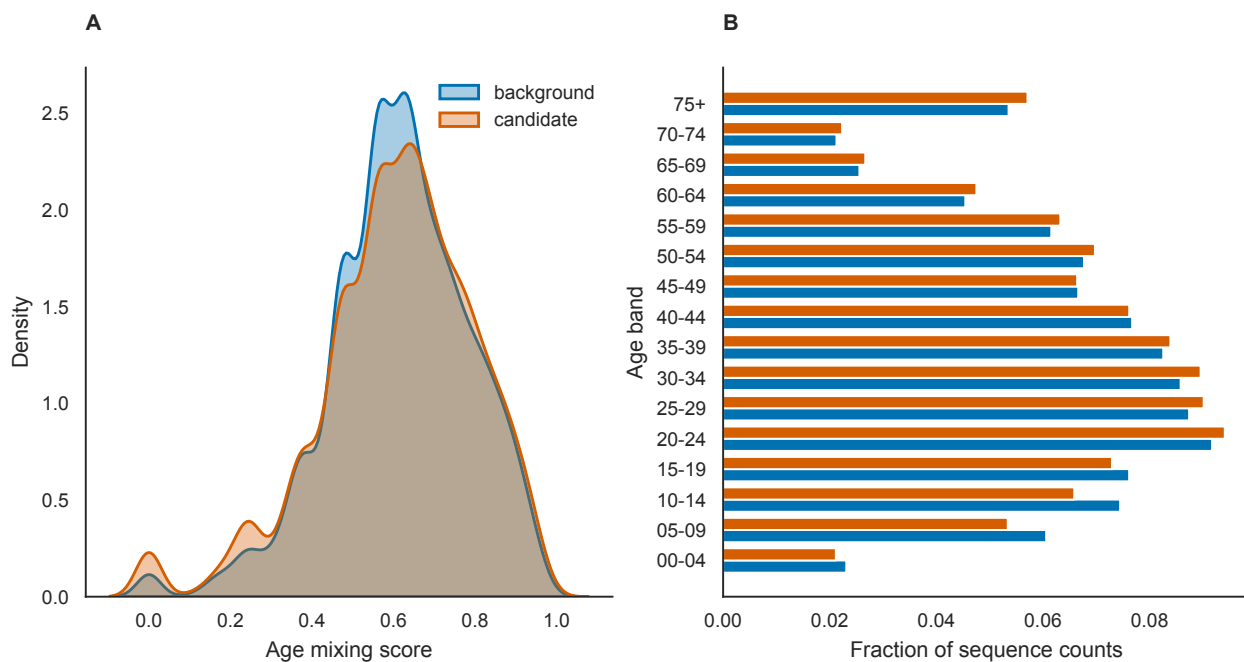
S5 Fig. Policy-era context for candidate SSE nodes. Descriptive candidate frequency and category composition across policy eras.



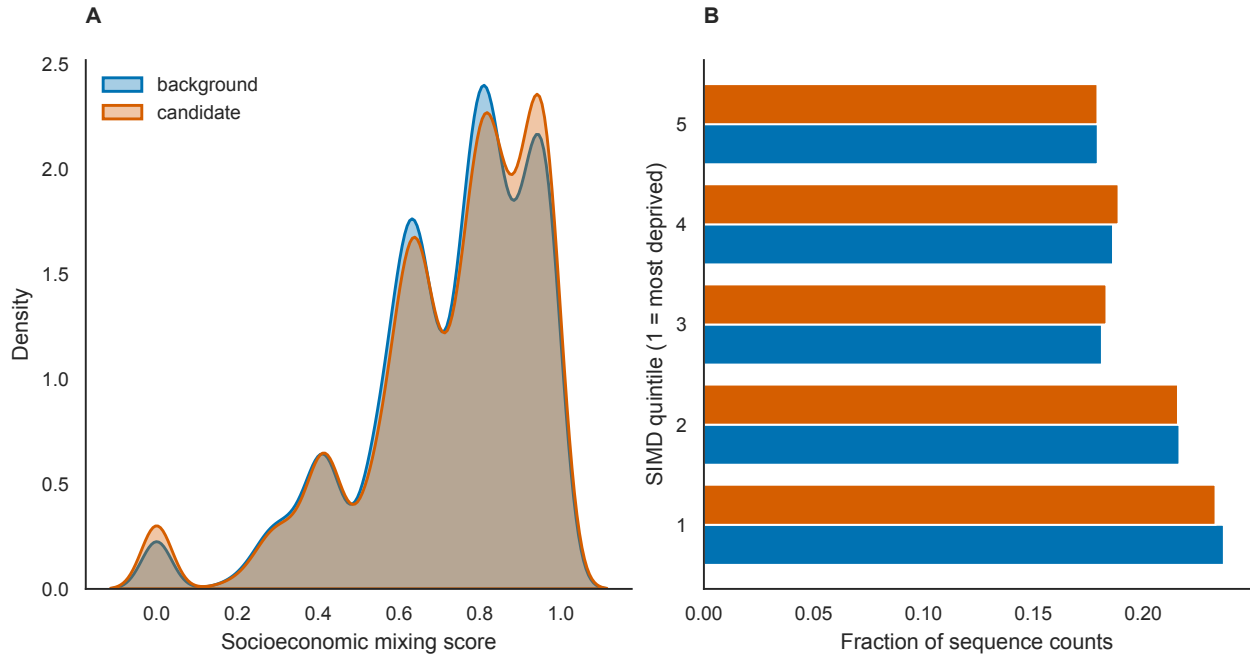
S6 Fig. Vaccination context for candidate SSE nodes. Vaccination summaries and age-conditional vaccination-mixing context for candidate and background nodes.



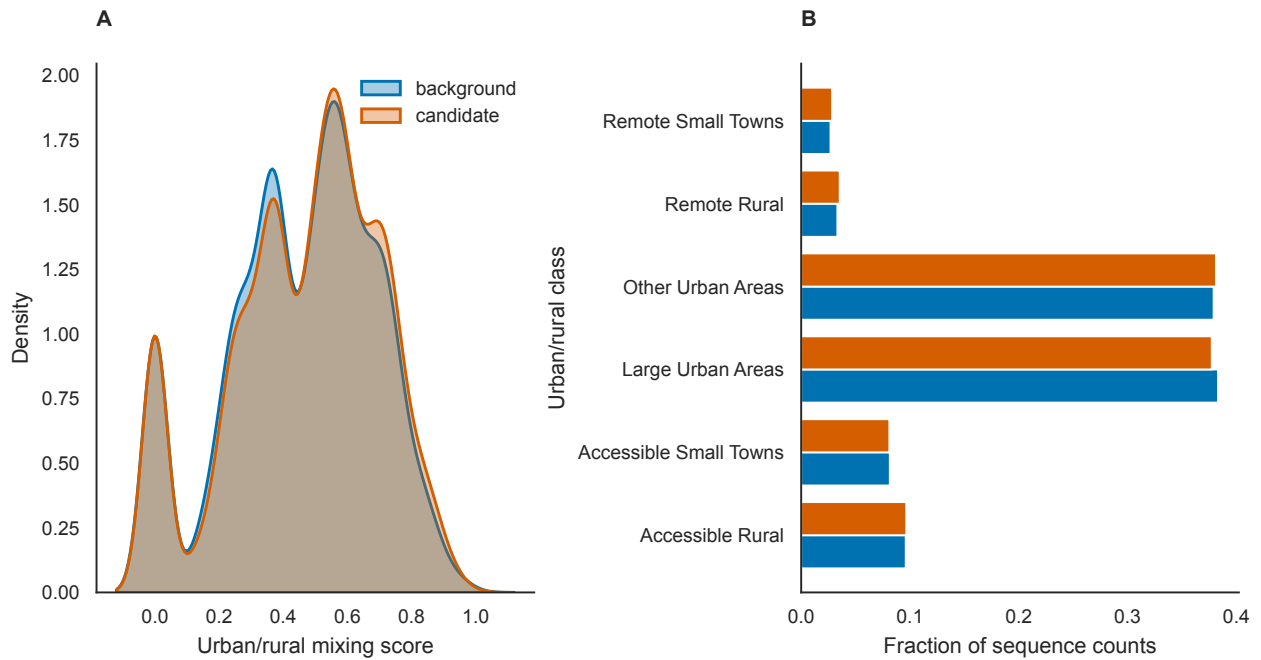
S7 Fig. Sex mixing and composition breakdown. Candidate/background sex mixing-score distributions paired with observed sex composition.



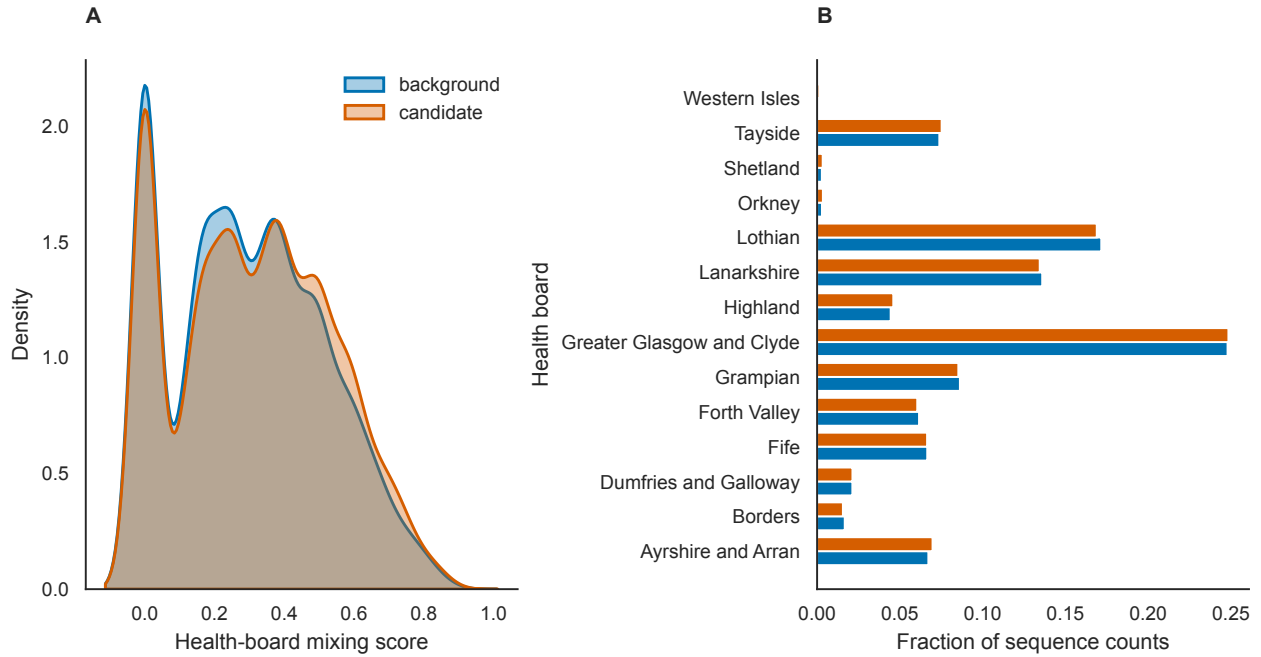
S8 Fig. Age mixing and composition breakdown. Candidate/background age mixing-score distributions paired with observed age-band composition.



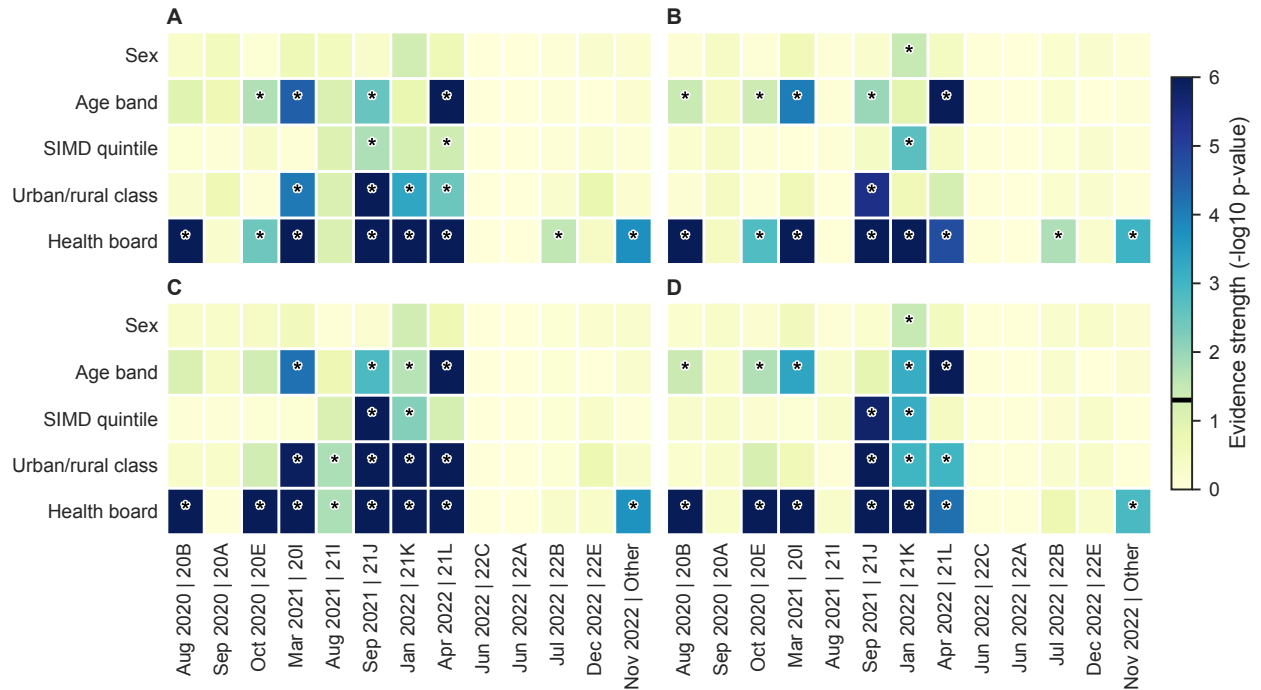
S9 Fig. SIMD mixing and composition breakdown. Candidate/background socioeconomic mixing-score distributions paired with observed SIMD-quintile composition.



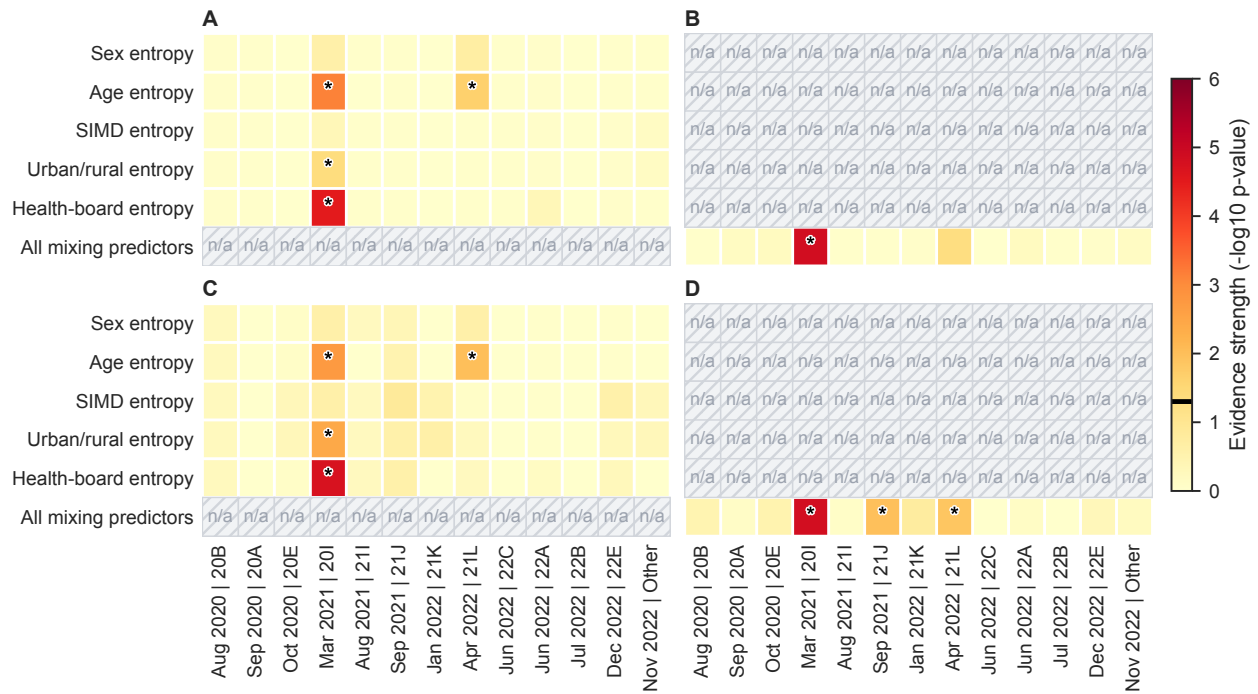
S10 Fig. Urban/rural mixing and composition breakdown. Candidate/background urban/rural mixing-score distributions paired with observed urban/rural composition.



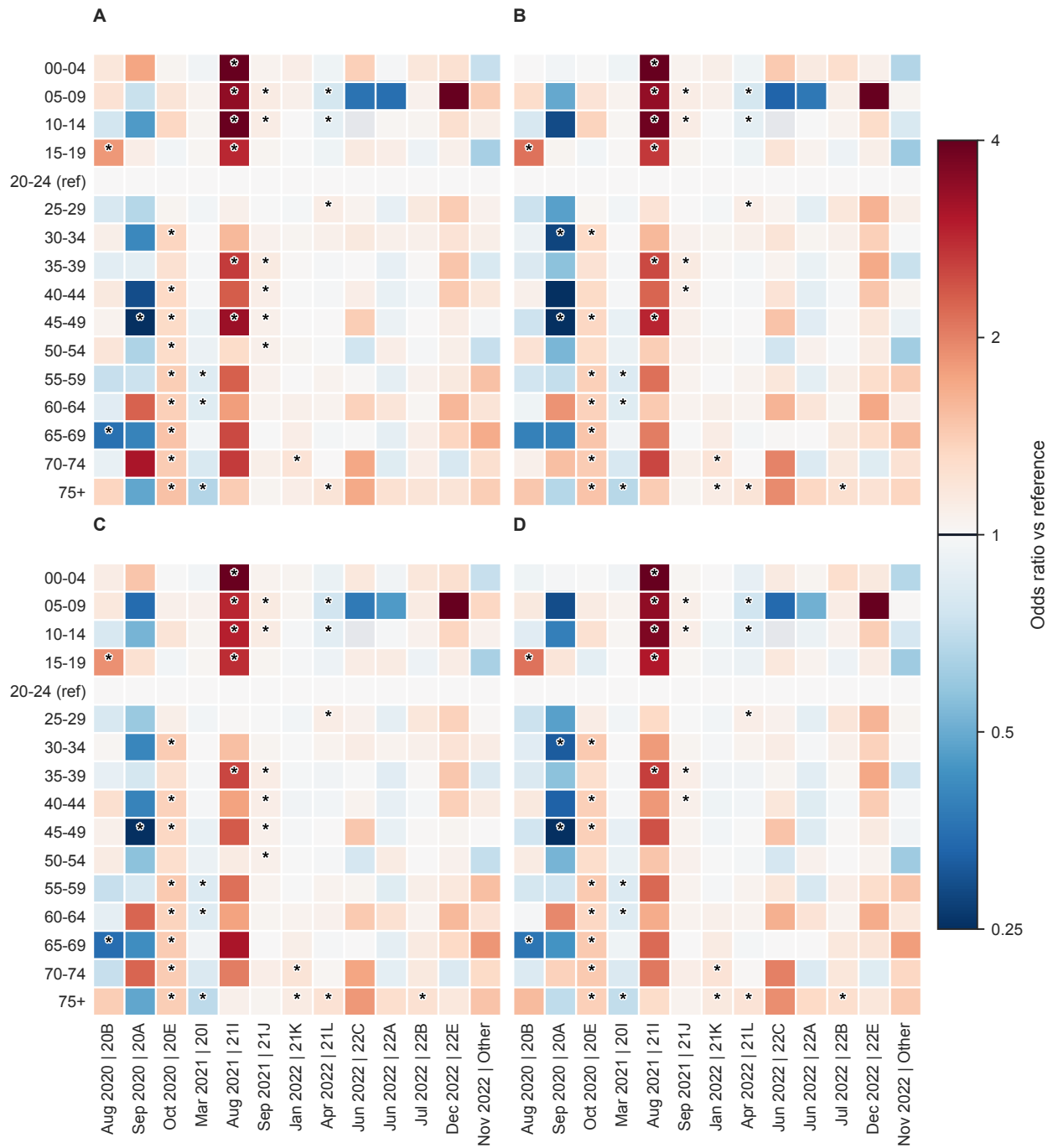
S11 Fig. Health-board mixing and composition breakdown. Candidate/background health-board mixing-score distributions paired with observed health-board composition.



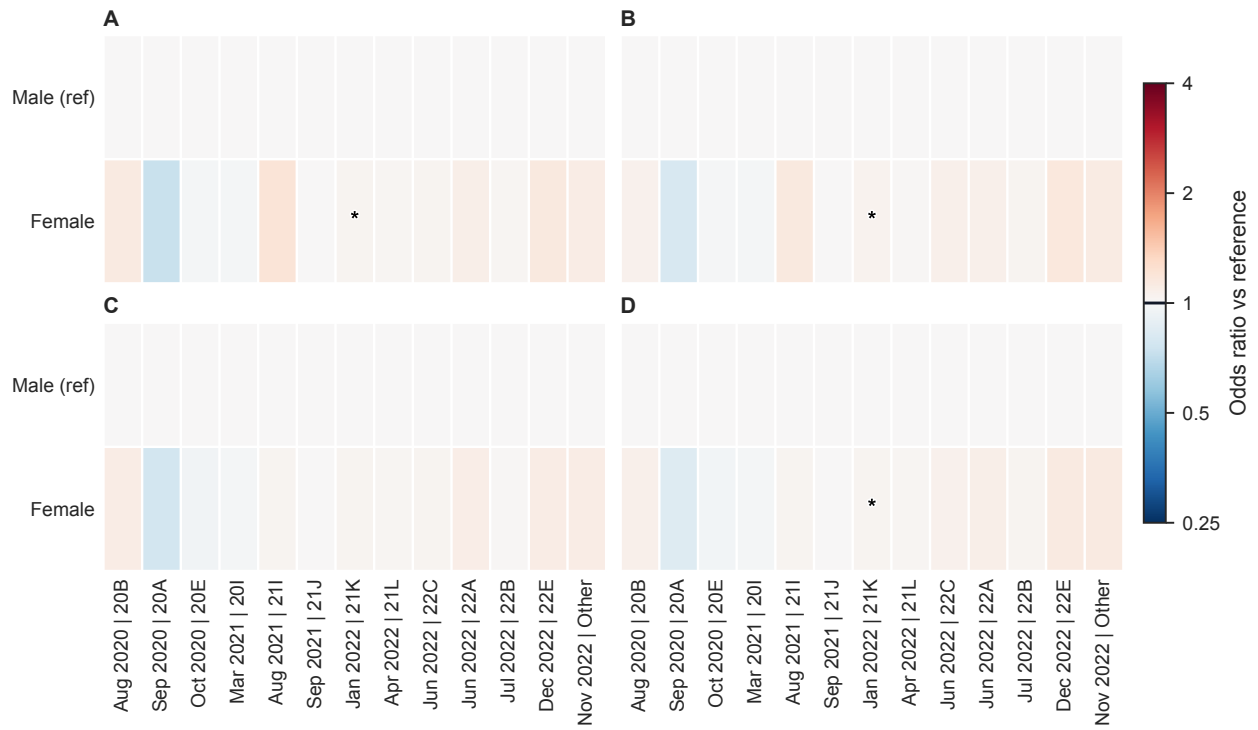
S12 Fig. Clade-stratified composition significance. Wald significance heatmap for composition predictors within major clade groups.



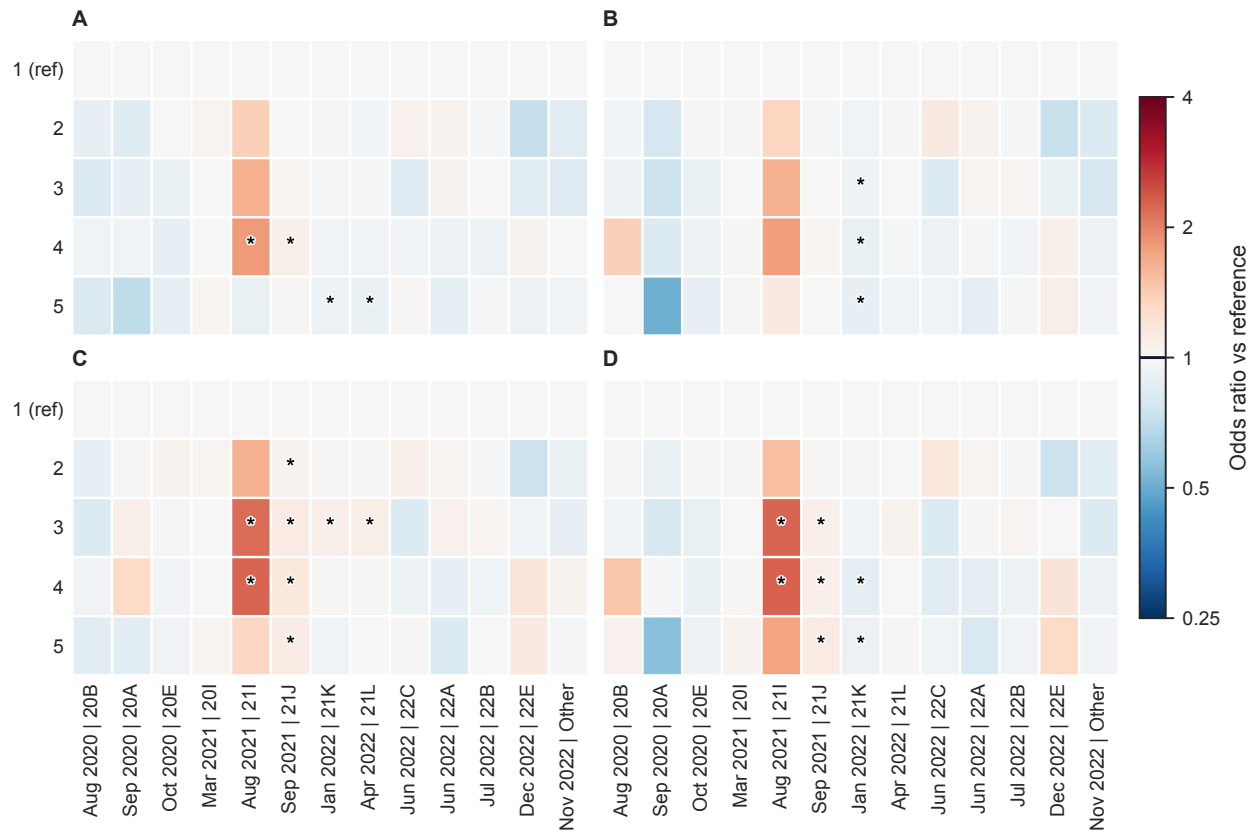
S13 Fig. Clade-stratified node-mixing significance. Wald significance heatmap for node-mixing predictors within major clade groups.



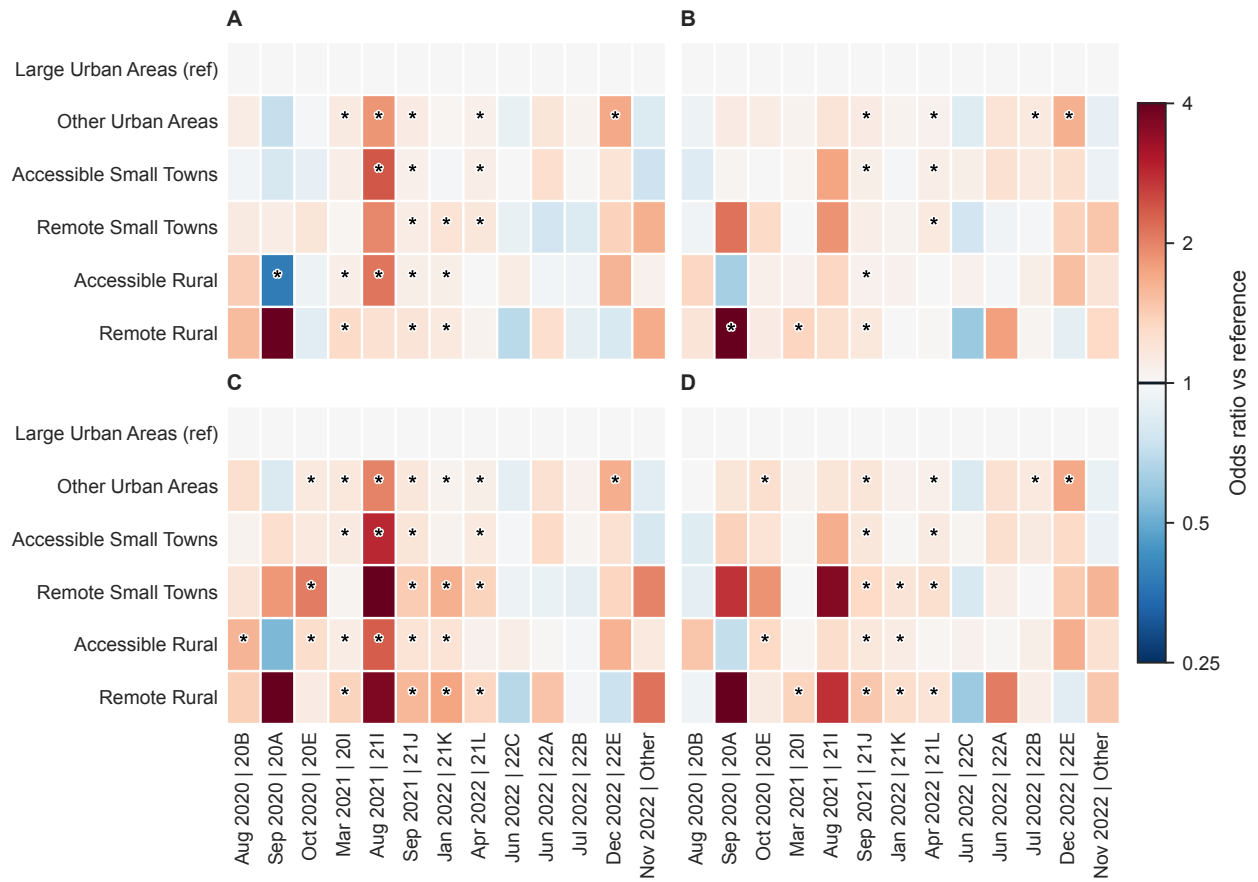
S14 Fig. Clade-stratified age-band odds ratios. Per-category age-band odds-ratio heatmap within clade groups.



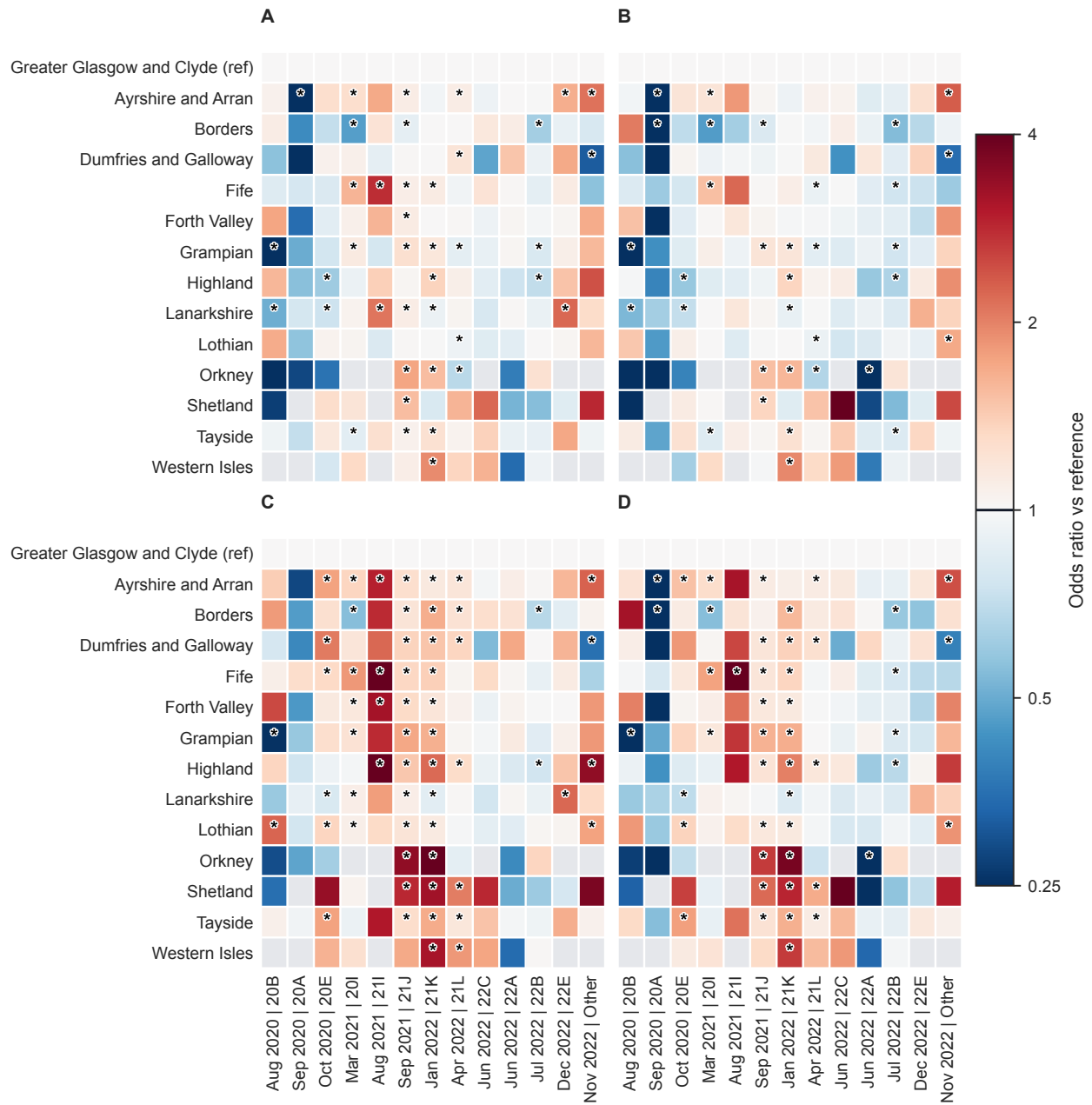
S15 Fig. Clade-stratified sex odds ratios. Per-category sex odds-ratio heatmap within clade groups.



S16 Fig. Clade-stratified SIMD-quintile odds ratios. Per-category SIMD-quintile odds-ratio heatmap within clade groups.



S17 Fig. Clade-stratified urban/rural odds ratios. Per-category urban/rural-class odds-ratio heatmap within clade groups.



S18 Fig. Clade-stratified health-board odds ratios. Per-category health-board odds-ratio heatmap within clade groups.

Supporting tables

S1 Table. Composition Wald tests. Omnibus Wald tests for sequence-composition predictors of candidate-node membership, by model set and specification.

Model set	Specification	Predictor	Reference/scale	df	Wald chi-square	FDR-adjusted P value	Sequences	Nodes
Expanded	Joint	Sex	Male	1	2.24	0.135	188,091	13,058
Expanded	Joint	Age band	20-24	15	23.68	0.089	188,091	13,058
Expanded	Joint	Simd quintile	1	4	8.66	0.089	188,091	13,058
Expanded	Joint	Urban/rural class	Large Urban Areas	5	135.00	<0.001	188,091	13,058
Expanded	Joint	Health board	Greater Glasgow and Clyde	13	371.63	<0.001	188,091	13,058
Expanded	Single predictor	Sex	Male	1	1.85	0.174	188,091	13,058
Expanded	Single predictor	Age band	20-24	15	24.69	0.068	188,091	13,058
Expanded	Single predictor	Simd quintile	1	4	33.73	<0.001	188,091	13,058
Expanded	Single predictor	Urban/rural class	Large Urban Areas	5	375.66	<0.001	188,091	13,058
Expanded	Single predictor	Health board	Greater Glasgow and Clyde	13	608.44	<0.001	188,091	13,058
Primary	Joint	Sex	Male	1	2.56	0.137	188,104	13,058
Primary	Joint	Age band	20-24	15	22.63	0.137	188,104	13,058
Primary	Joint	Simd quintile	1	4	3.08	0.544	188,104	13,058
Primary	Joint	Urban/rural class	Large Urban Areas	5	47.95	<0.001	188,104	13,058
Primary	Joint	Health board	Greater Glasgow and Clyde	13	78.38	<0.001	188,104	13,058
Primary	Single predictor	Sex	Male	1	2.44	0.148	188,104	13,058
Primary	Single predictor	Age band	20-24	15	22.74	0.148	188,104	13,058
Primary	Single predictor	Simd quintile	1	4	6.25	0.181	188,104	13,058
Primary	Single predictor	Urban/rural class	Large Urban Areas	5	91.06	<0.001	188,104	13,058
Primary	Single predictor	Health board	Greater Glasgow and Clyde	13	119.20	<0.001	188,104	13,058

S2 Table. Composition odds ratios. Per-category adjusted odds ratios (95% CI) for all composition predictors across model sets and specifications (152 rows); provided as a machine-readable file: `table03_composition_odds_ratios.csv`

S3 Table. Composition model fit statistics. McFadden pseudo- R^2 , AIC, BIC, and likelihood summaries for composition models.

Model set	Specification	Predictor	McFadden pseudo- R^2	AIC	BIC	Log likelihood	Sequences	Nodes	Converged
Expanded	Joint	All composition predictors		333429.9		-166587.0	188,091	13,058	True
Expanded	Single predictor	Sex		334141.8		-166979.9	188,091	13,058	True
Expanded	Single predictor	Age band		334146.9		-166968.4	188,091	13,058	True
Expanded	Single predictor	Simd quintile		334115.8		-166963.9	188,091	13,058	True
Expanded	Single predictor	Urban/rural class		333773.7		-166791.8	188,091	13,058	True
Expanded	Single predictor	Health board		333553.0		-166673.5	188,091	13,058	True
Primary	Joint	All composition predictors		335081.2		-167416.6	188,104	13,058	True
Primary	Single predictor	Sex		335202.5		-167514.2	188,104	13,058	True
Primary	Single predictor	Age band		335210.1		-167504.1	188,104	13,058	True
Primary	Single predictor	Simd quintile		335204.7		-167512.3	188,104	13,058	True
Primary	Single predictor	Urban/rural class		335121.7		-167469.9	188,104	13,058	True
Primary	Single predictor	Health board		335109.0		-167455.5	188,104	13,058	True

S4 Table. Node-mixing Wald tests. Omnibus Wald tests for entropy-based node-mixing predictors.

Model set	Specification	Predictor	Reference/scale	df	Wald chi-square	FDR-adjusted P value	Nodes
Expanded	Joint	All mixing predictors	per 1 null-model SD increase in entropy	5	26.20	<0.001	12,966
Expanded	Single predictor	Sex entropy z	per 1 null-model SD increase in entropy	1	1.05	0.305	12,966
Expanded	Single predictor	Age entropy z	per 1 null-model SD increase in entropy	1	11.62	0.003	12,966
Expanded	Single predictor	Simd entropy z	per 1 null-model SD increase in entropy	1	3.97	0.077	12,966
Expanded	Single predictor	Urban rural entropy z	per 1 null-model SD increase in entropy	1	2.50	0.142	12,966
Expanded	Single predictor	Health board entropy z	per 1 null-model SD increase in entropy	1	5.51	0.047	12,966
Primary	Joint	All mixing predictors	per 1 null-model SD increase in entropy	5	10.24	0.069	12,966
Primary	Single predictor	Sex entropy z	per 1 null-model SD increase in entropy	1	1.65	0.285	12,966
Primary	Single predictor	Age entropy z	per 1 null-model SD increase in entropy	1	8.03	0.023	12,966
Primary	Single predictor	Simd entropy z	per 1 null-model SD increase in entropy	1	1.45	0.285	12,966
Primary	Single predictor	Urban rural entropy z	per 1 null-model SD increase in entropy	1	0.00	0.948	12,966
Primary	Single predictor	Health board entropy z	per 1 null-model SD increase in entropy	1	1.95	0.285	12,966

S5 Table. Node-mixing odds ratios. Adjusted odds ratios (per null-model SD) for node-mixing entropy predictors.

Model set	Specification	Predictor	Contrast	Reference/scale	Odds ratio (95% CI)	P value	Nodes
Expanded	Joint	Sex entropy z-score	Sex entropy z-score	per 1 null-model SD increase in entropy	0.99 (0.97-1.02)	0.632	12,966
Expanded	Joint	Age entropy z-score	Age entropy z-score	per 1 null-model SD increase in entropy	0.97 (0.95-0.99)	0.004	12,966
Expanded	Joint	SIMD entropy z-score	SIMD entropy z-score	per 1 null-model SD increase in entropy	0.99 (0.97-1.01)	0.243	12,966
Expanded	Joint	Urban/rural entropy z-score	Urban/rural entropy z-score	per 1 null-model SD increase in entropy	1.04 (1.02-1.07)	<0.001	12,966
Expanded	Joint	Health-board entropy z-score	Health-board entropy z-score	per 1 null-model SD increase in entropy	0.98 (0.96-1.00)	0.010	12,966
Expanded	Single predictor	Sex entropy z	Sex entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.99 (0.96-1.01)	0.305	12,966
Expanded	Single predictor	Age entropy z	Age entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.96 (0.94-0.98)	<0.001	12,966
Expanded	Single predictor	Simd entropy z	SIMD entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.98 (0.96-1.00)	0.046	12,966
Expanded	Single predictor	Urban rural entropy z	Urban/rural entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	1.02 (1.00-1.04)	0.114	12,966
Expanded	Single predictor	Health board entropy z	Health-board entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.98 (0.97-1.00)	0.019	12,966
Primary	Joint	Sex entropy z-score	Sex entropy z-score	per 1 null-model SD increase in entropy	0.99 (0.96-1.01)	0.361	12,966
Primary	Joint	Age entropy z-score	Age entropy z-score	per 1 null-model SD increase in entropy	0.97 (0.95-0.99)	0.015	12,966
Primary	Joint	SIMD entropy z-score	SIMD entropy z-score	per 1 null-model SD increase in entropy	1.00 (0.98-1.02)	0.690	12,966
Primary	Joint	Urban/rural entropy z-score	Urban/rural entropy z-score	per 1 null-model SD increase in entropy	1.01 (0.99-1.04)	0.407	12,966
Primary	Joint	Health-board entropy z-score	Health-board entropy z-score	per 1 null-model SD increase in entropy	0.99 (0.98-1.01)	0.359	12,966
Primary	Single predictor	Sex entropy z	Sex entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.98 (0.96-1.01)	0.199	12,966
Primary	Single predictor	Age entropy z	Age entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.97 (0.95-0.99)	0.005	12,966
Primary	Single predictor	Simd entropy z	SIMD entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.99 (0.97-1.01)	0.228	12,966
Primary	Single predictor	Urban rural entropy z	Urban/rural entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	1.00 (0.98-1.02)	0.948	12,966
Primary	Single predictor	Health board entropy z	Health-board entropy z-score vs per 1 null-model SD increase in entropy	per 1 null-model SD increase in entropy	0.99 (0.98-1.00)	0.162	12,966

S6 Table. Node-mixing model fit statistics. Fit statistics for node-mixing models.

Model set	Specification	Predictor	McFadden pseudo-R2	AIC	BIC	Log likelihood	Nodes	Converged
Expanded	Joint	All mixing predictors		16957.8		-8383.9	12,966	True
Expanded	Single predictor	Sex entropy z		16975.7		-8396.9	12,966	True
Expanded	Single predictor	Age entropy z		16964.9		-8391.4	12,966	True
Expanded	Single predictor	Simd entropy z		16972.8		-8395.4	12,966	True
Expanded	Single predictor	Urban rural entropy z		16974.3		-8396.2	12,966	True
Expanded	Single predictor	Health board entropy z		16971.2		-8394.6	12,966	True
Primary	Joint	All mixing predictors		17083.9		-8450.9	12,966	True
Primary	Single predictor	Sex entropy z		17084.7		-8455.4	12,966	True
Primary	Single predictor	Age entropy z		17078.3		-8452.1	12,966	True
Primary	Single predictor	Simd entropy z		17085.0		-8455.5	12,966	True
Primary	Single predictor	Urban rural entropy z		17086.5		-8456.2	12,966	True
Primary	Single predictor	Health board entropy z		17084.5		-8455.3	12,966	True

The following clade-stratified result tables are large and are provided as machine-readable comma-separated-value files in the `sse_detection/manuscript/tables/` directory; the corresponding figures (S12–S18 Fig) summarise them visually.

- **S7 Table. Clade-stratified composition Wald summary.** `table08_clade_composition_wald_summary.csv`
- **S8 Table. Clade-stratified node-mixing Wald summary.** `table09_clade_mixing_wald_summary.csv`
- **S9 Table. Clade-stratified age-band odds ratios.** `table10_clade_age_band_or_summary.csv`
- **S10 Table. Clade-stratified sex odds ratios.** `table11_clade_sex_or_summary.csv`
- **S11 Table. Clade-stratified SIMD-quintile odds ratios.** `table12_clade_simd_or_summary.csv`
- **S12 Table. Clade-stratified urban/rural odds ratios.** `table13_clade_urban_rural_or_summary.csv`
- **S13 Table. Clade-stratified health-board odds ratios.** `table14_clade_health_board_or_summary.csv`